

NISHCHAL MARUR

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SUMMARY

Machine Learning Engineer with 4+ years of experience delivering end-to-end Production ML systems. Engineered high-stakes authentication pipelines for luxury goods at 96% TPR, developed real-time recommendation engines and Agentic RAG pipelines, while compressing edge models by 10x for scalable mobile deployment.

EXPERIENCE

AI Engineer Intern | Connyct Inc | New York Nov 2025 - Jan 2026

- Built a Two-Tower hybrid recommendation engine combining sentence transformers and Elasticsearch with multi-signal user profiling, achieving Recall@10 of 0.85 for personalized events at <40 ms latency.
- Implemented MCP tools for the LLM orchestrator with location-aware reranking, composite relevance scoring, and Redis caching, to continuously optimize CTR with A/B testing framework.

Research Assistant | PRG Lab (PI: Cornelia Fermuller) | Maryland Jun 2025 - Aug 2025

- Developed a cross-modal generation pipeline converting performer pose sequences to violin audio using DDSP, Transformer encoders, and autoencoder-based MIDI synthesis, enabling audio generation directly from motion capture.

Machine Learning Engineer II | Entrupy Inc | Bangalore Aug 2021 - Aug 2024

- Built a deep learning authentication pipeline automating 95% of the verification volume and reducing manual expert reviews, achieving 96% TPR at 5% FPR across 50K+ monthly items.
- Led the end-to-end R&D of a 3D document unwarping system that outperformed SOTA methods, resulting in 0.84 SSIM score and 23% increase in OCR accuracy, trained exclusively on synthetic data.
- Improved on-device segmentation IoU from 87% to 96% using SAM-HQ and trained a compact EfficientNet for CoreML deployment, achieving 10x model compression with FP16 quantization for real-time auto-capture on iOS.
- Optimized macro fingerprinting pipeline for luxury goods tracking using patch embedding similarity, boosting TPR by 15% and reducing authentication latency from 25s to under 15s with Ray parallel processing.
- Designed synthetic data pipelines using Blender Python and Stable Diffusion inpainting to simulate camera intrinsics, lighting, and material textures, reducing time-to-production for new models from weeks to days.

MLOps Intern | IBM | Bangalore Jan 2021 - Jul 2021

- Reduced inference latency by 15ms in IBM Watson Cloud deployments by optimizing batch prediction pipelines using Go concurrency and chunked downloads on Kubernetes, and contributed to the design of a new internal inference architecture.

Software Engineering Intern | SLK Software | Bangalore May 2020 - Jul 2020

- Saved developers 10+ hours weekly in debugging time by building a centralized log aggregation system using ELK Stack, Filebeat, Node.js across 5+ distributed components.

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL, Scala, Node.js, Bash/Shell

Frameworks: PyTorch, TensorFlow, HuggingFace, TensorRT, ONNX, CoreML, SAM, OpenCV, Scikit-Learn, Pandas, NumPy

GenAI: LangChain, LangGraph, LlamaIndex, vLLM, LoRA/Fine-Tuning, MCP, Stable Diffusion

Infra & MLOps: AWS (EC2, S3, SageMaker), Docker, Kubernetes, Ray, MLflow, Kubeflow, Git, FastAPI, CI/CD

Data Systems: Elasticsearch, Pinecone, Redis, PostgreSQL, MongoDB, PySpark, Kafka, Blender

ML Domains: Computer Vision, Multimodal AI, Large Language Models, Recommendation Systems, Edge AI, Model Optimization, Agentic RAG, Reinforcement Learning, Synthetic Data Generation

PROJECTS

CAFBrain: Multimodal LLM Platform (Agentic RAG | LLM)

- Built an agentic RAG system using LangGraph with multi-step tool routing and autonomous query decomposition across 5000+ multimodal documents, reducing Capital Area Food Bank's grant proposal creation from hours to under a minute.

Temporal Change Retrieval (Computer Vision | Multimodal)

- Developed a dual-encoder architecture with a custom difference-attention module and LoRA-adapted RemoteCLIP for temporal vision-language alignment, achieving 64% Recall@10 on natural-language change queries.

In-Context Learning for Autonomous Drone Navigation (Robotics | Transformers)

- Enabled zero-retraining adaptation to unseen drone racing environments from just 3 demos via cross-attention over context embeddings, achieving 42% lower MSE than PPO baselines across 100+ procedurally generated tracks.

EDUCATION

Master of Science (M.S) in Machine Learning Aug 2024 - May 2026

University of Maryland, College Park

GPA: 3.8/4

Bachelor of Technology (B.Tech) in Computer Science Aug 2017 - May 2021

PES University, Bangalore

GPA: 3.6/4